Déjà Vu, Strange Attractors, and Orphaned Topological Caches: A Dynamical-Systems Model of Timeline Bifurcation and Collective Memory Divergence

Nickolas P. J. Schoff
Independent Researcher
Schoff Research Program
Salem / Mason, New Hampshire, United States

with conceptual contribution by Grok (xAI)

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Abstract

This paper proposes a dynamical-systems model of déjà vu and the Mandela Effect grounded in the concepts of strange attractors, topological cache pointers, and timeline bifurcation. Standard cognitive accounts treat déjà vu as a familiarity–novelty conflict within recognition memory and the Mandela Effect as a collective false-memory phenomenon driven by schema, source monitoring, and social reinforcement. The present framework accepts those proximate mechanisms while relocating their ultimate structural cause to the navigation of high-dimensional possibility space. Strange attractors—dense, high-probability future configurations—exert gravitational pull on invariant agencies. Orthogonal pings into this space can pre-cache partial renders of attractor coordinates. When the rendered consensus trajectory bifurcates or settles into an adjacent basin, previously written cache pointers become orphaned. The resulting mismatch is experienced individually as deep-state déjà vu and, when sufficiently widespread, as collective Mandela-type divergence. The model draws on attractor dynamics in consciousness and memory research, dual-process recognition theory, and the topological architectures developed in the accompanying papers of this series. Speculative elements concerning Dimension-W navigation and non-local cache writes are clearly demarcated.

Keywords: déjà vu, Mandela Effect, strange attractors, topological cache, timeline bifurcation, invariant agency, dynamical systems, collective false memory

Introduction

Déjà vu is the sudden, often intense impression that a current situation has been experienced before, accompanied by the metacognitive awareness that the impression is false or unexplained. Contemporary cognitive neuroscience frames it as a conflict between a familiarity signal (commonly associated with perirhinal cortex) and a novelty or contextual signal (associated with hippocampal processing), producing a momentary failure of source monitoring or expectation checking (Martin et al.; Aitken, Jentzsch, & O'Connor). Laboratory work has further linked the

experience to an illusory sense of precognition: the feeling that one is about to know what happens next, even though actual predictive accuracy remains at chance (Cleary and colleagues).

The Mandela Effect designates a class of collective false memories in which large numbers of people confidently recall the same non-canonical detail of a public fact, logo, or historical event. Empirical studies confirm the phenomenon for certain visual icons and attribute it to schema-driven reconstruction, exposure to inaccurate variants, source confusion, and social reinforcement (Prasad & Bainbridge; related cognitive literature). No single proximate mechanism accounts for all instances.

The present paper accepts these proximate accounts while offering a higher-order dynamical interpretation. Drawing on attractor models of consciousness and memory, and on the topological architectures formalized in earlier papers of this series, it treats deep-state déjà vu and Mandela-type divergences as signatures of orphaned topological cache pointers left by navigation among strange attractors in a high-dimensional possibility space.

Strange Attractors in Possibility Space

In dynamical systems theory, an attractor is a set of states toward which a system evolves from a range of initial conditions. Strange attractors are characterized by fractal structure and sensitive dependence on initial conditions. Attractor-network models have long been invoked in theories of consciousness and memory: stable or metastable attractor states are proposed as the computational correlates of conscious contents, and memory retrieval is modeled as trajectory convergence within multi-vector dynamical landscapes (Grossberg; Rumelhart et al.; more recent multi-vector and criticality approaches).

Within the present framework, the unrendered future is not a uniform distribution of noise. Certain topological configurations possess high gravitational weight—dense basins that pull nearby trajectories. These are treated as strange attractors in possibility space. An invariant agency

does not wander randomly; it navigates under the influence of these basins. A timeline shift corresponds to the breaking of orbit from one local attractor and capture by another.

Topological Caching and the Phenomenology of Déjà Vu

An orthogonal ping into possibility space can return a partial render of a distant attractor and write it to a local topological cache as a target waypoint, even though the corresponding physical frames have not yet been traversed. Subsequent micro-adjustments of constraint and action gradually steer the rendered trajectory into the new basin. The moment of physical handshake—the arrival of the local world-line at the pre-cached coordinates—produces a sudden, high-intensity familiarity signal whose source cannot be located in ordinary episodic memory. The resulting phenomenology is deep-state déjà vu: not a short-range cache glitch, but confirmation that an invariant agency has entered a new gravitational orbit.

This account is consistent with laboratory findings that déjà vu is accompanied by an illusory predictive feeling. The pre-cached waypoint creates the subjective sense that the unfolding sequence is already known; actual predictive accuracy remains low because the cache contains geometric or topological coordinates rather than detailed episodic content.

Bifurcation, Orphaned Pointers, and the Mandela Effect

Macro-attractors are rarely simple funnels. Because possibility space contains dense fractal structure, a high-probability basin often features a topological bifurcation—two nearly identical, nearly equiprobable adjacent wells. Collective navigational thrust may split across the two branches. At the moment of consensus collapse, the rendered 3D trajectory settles into one branch; the other becomes an orphaned branch that remains uncollapsed in the higher-dimensional latency.

Invariant agencies that pre-cached the orphaned branch retain local topological pointers to coordinates that were never rendered in the consensus timeline. When those individuals later attempt to recall the corresponding public fact or icon, the pointer returns a coherent but non-canonical detail. Because many agencies may have cached the same adjacent branch, the mismatch

appears collective. Proximate cognitive mechanisms—schema completion, source monitoring failure, social reinforcement—then amplify and stabilize the divergent memory. The Mandela Effect is thereby interpreted as the macroscopic signature of widespread orphaned topological cache pointers following bifurcation of a macro-attractor.

Discussion and Boundaries

The model does not deny the well-documented cognitive mechanisms of false memory and familiarity–novelty conflict. It relocates their structural origin. Ordinary déjà vu may still be a short-range recognition glitch; deep-state déjà vu and systematic collective divergences are proposed as consequences of attractor navigation and cache orphaning in a higher-dimensional possibility space. The speculative components—Dimension-W as the ambient possibility field, orthogonal pings as non-local cache writes, and the precise topology of bifurcation—are program-internal hypotheses continuous with the Castor–Pollux and continuity architectures developed earlier. They remain open to empirical constraint and formal refinement.

Conclusion

Déjà vu and the Mandela Effect can be modeled as dynamical consequences of navigation among strange attractors in possibility space. Pre-caching of attractor coordinates produces topological pointers that, after bifurcation or orbit shift, become orphaned relative to the rendered consensus trajectory. The resulting mismatch is experienced as profound familiarity without episodic source (deep-state déjà vu) or as collective confident divergence from canonical public memory (Mandela Effect). The framework unifies attractor dynamics, recognition-memory conflict, and topological continuity under a single generative hypothesis while preserving a clear boundary between established cognitive science and theoretical extension.

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